

Rainfall Patterns and 30-Minute Rainfall Prediction with Apache Spark

DSA5208 | Scalable Distributed Computing, Project 2

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Abstract: We use Apache Spark to analyse eight years (2017–2024) of historical rainfall data from Singapore, comprising 48.5 million five-minute observations across 91 monitoring stations. The study focuses on two objectives: exploring how rainfall patterns vary within and across years, and developing a machine learning model to estimate the probability of rainfall exceeding a specified threshold at a given station in the next 30 minutes. Following data cleaning and quality checks, we developed a logistic regression baseline and a random forest model using features such as recent rainfall and observations from nearby stations. The models were evaluated on unseen data from 2024, with the random forest achieving a PR-AUC of 0.465 and ROC-AUC of 0.969 for a 5 mm rainfall threshold, outperforming the baseline at every evaluated thresholds. The complete analysis was executed both locally and on Google Cloud Dataproc, producing consistent results across environments. On Dataproc, doubling the number of worker nodes reduced the execution time of the main Spark processing job by approximately 50%, demonstrating the scalability of the approach.

1. Executive Summary

This project uses Apache Spark to process and analyse eight years (2017–2024) of historical rainfall data from Singapore, comprising 48.5 million five-minute observations across 91 monitoring stations (approximately 7.5 GB of CSV data). The analysis covers the complete workflow, including data cleaning, exploratory analysis, feature engineering, model development and evaluation, using Spark for large-scale data processing and MLlib for machine learning.

Task 1: Rainfall Pattern Analysis. The analysis reveals distinct seasonal and daily rainfall patterns, significant spatial differences across Singapore, and considerable year-to-year variability. Rainfall generally follows two wetter seasonal periods, exhibits different patterns throughout the day, and tends to be higher in western Singapore. Differences in annual rainfall are largely associated with changes in rainfall frequency rather than intensity.

Task 2: Rainfall Prediction. We developed a machine learning model to estimate the probability that a monitoring station will record rainfall exceeding a specified threshold within the next 30 minutes. A logistic regression model was established as a baseline, followed by a random forest model incorporating additional features such as recent rainfall observations, dry-spell duration and rainfall at nearby stations. The final model was selected using year-blocked cross-validation and evaluated on unseen data from 2024. The random forest generally outperformed the logistic regression baseline, particularly in identifying rainfall events and producing reliable probability estimates.

Model	Mean PR-AUC	Mean ROC-AUC	Mean log loss	PR-AUC at 5 mm	ROC-AUC at 5 mm
Final model: random forest (Stage 2)	0.450	0.973	0.0266	0.459	0.970
Stage 1: logistic regression	0.408	0.921	0.0328	0.418	0.912
Climatology baseline	0.023	0.695	0.0553	0.018	0.684

Table 1. Final model against the Stage 1 model and the climatology baseline on identical 2024 rows (1,255,003 rows per threshold; means over the 1, 2, 5, 10 and 20 mm thresholds).

Scalability Assessment. The complete analysis was executed both locally and on Google Cloud Dataproc, producing consistent results across environments. On Dataproc, doubling the number of worker nodes reduced the execution time of the main Spark processing job by approximately 50%, highlighting the benefits of distributed processing for large-scale rainfall analysis.

Key Insight. Recent rainfall observations are the strongest predictors of rainfall in the next 30 minutes, particularly rainfall recorded at the station itself and at nearby monitoring stations. While seasonal and time-of-day features provide useful context, incorporating recent and neighbouring rainfall observations substantially improves predictive performance. This highlights the importance of capturing both temporal and spatial rainfall patterns when developing short-term rainfall prediction models.

2. Problem Statement & Success Metrics

Task 1. Analyse rainfall patterns within each year and across 2017–2024, using the rainfall record and visualisations.

Task 2. Build and evaluate a Spark MLlib model estimating the probability that a monitoring station accumulates more than a given amount of rain in the next 30 minutes. The inputs are the station, prediction time and threshold h.

2.1 Why accuracy is not a suitable evaluation metric

Only 0.83% of station-times exceed 5 mm in 30 minutes. Always predicting no heavy rainfall gives 99.17% accuracy while missing every event. We therefore evaluate event discrimination, probability quality and consistency across thresholds.

Evaluation Objective	Metric	What it measures	Desired Performance
Discrimination	PR-AUC	Measures how effectively the model identifies rare rainfall events across different probability thresholds.	Higher is better
	ROC-AUC	Measures the trade-off between precision and recall when identifying rare rainfall events.	Higher is better

Evaluation Objective	Metric	What it measures	Desired Performance
Probability Quality	Log loss	Accuracy of predicted probabilities, with greater penalties for confident but incorrect predictions.	Lower is better
	Brier skill score	Reduction in mean squared probability error relative to the climatology baseline.	Higher is better
	Reliability	Whether predicted probabilities match observed rainfall frequencies.	Closer to the diagonal is better.
Logical Consistency	Monotonicity	Whether predicted probabilities decrease or remain constant as rainfall thresholds increase.	Zero violations

Table 2. Evaluation metrics used throughout. All are computed on the 2023 validation year for model choice and once on the 2024 test year.

2.2 One model for every threshold

> **Design choice:** We use a single binary classification model with the rainfall threshold (h) as an input feature, rather than training separate models for each threshold. Each station-time observation is paired with multiple thresholds, with a label of 1 if rainfall in the next 30 minutes exceeds the threshold and 0 otherwise. This allows the model to estimate rainfall probabilities for different thresholds within the training range (0.2–60 mm).

3. Dataset & Computing Environment

The eight annual CSVs contain 7.5 GB of five-minute observations: **48.5 million records from 91 stations.** Between 54 and 80 stations report each year, reflecting network changes including the 2020 expansion.

Column group	Columns
Time	timestamp (ISO 8601, +08:00), update_timestamp, reading_update_timestamp, date
Station	station_id, station_name, station_device_id, location_longitude, location_latitude
Measurement	reading_value (mm per 5 minutes), reading_type, reading_unit

Table 3. Dataset schema. 12 columns, one row per station per 5-minute interval.

3.1 Computing environment

The submitted run uses Apache Spark in local mode on one laptop. We also timed the core Spark job on Google Cloud Dataproc with 2 and 4 workers to measure horizontal scaling (Section 8.5). The complete notebook ran on a 2-worker cluster using Spark on YARN and HDFS storage (Section 8.6).

Item	Value
Machine	Apple MacBook Pro, Apple M4 Max, 16 cores (12 performance + 4 efficiency), 128 GB RAM, macOS
Spark	Apache Spark 3.5.5 (PySpark), master local[14], driver memory 16g, 96 shuffle partitions
Software	Python 3.11.17, OpenJDK 17, pandas / NumPy / Matplotlib 3.0.6 / 1.26.4 / 3.11.2
Storage layers	CSV 7.5 GB → bronze Parquet 49 MB (typed raw copy) → silver 413 MB (clean readings) → gold 5.9 GB (features, training sets, predictions); Snappy-compressed, partitioned by year
Saved models	final random forest 11.5 MB; Stage 1 logistic baseline 13 KB
Cloud: scaling runs	Google Cloud Dataproc 2.2 (Debian 12, Spark 3.5), region asia-southeast1 (Singapore); master and workers n2-standard-2 (2 vCPU, 8 GB), 50 GB standard disks; 2 and then 4 workers; data in a Cloud Storage bucket; gcloud dataproc jobs submit pyspark (cloud/dataproc_benchmark.py)
Cloud: complete run	Dataproc 2.2, master + 2 workers n2-highmem-4 (4 vCPU, 32 GB, Intel Xeon 2.8 GHz), 100/75/75 GB balanced persistent disks (the education project's quotas: 12 vCPUs, 250 GB SSD); Spark 3.5.3 on YARN, data layers on HDFS, driver 16 GB; Cloud NAT for package installation

Table 4. Computing environment: the laptop as recorded by the notebook (t00_environment.csv), the cluster as configured for the Dataproc jobs.

Stage	Minutes
Ingestion and data-quality audit	0.4
Cleaning and pre-processing	0.6
Task 1 analyses	0.5
Stage 1 features, sets, baselines and models	7.9
Stage 2 features and feature screening	1.2
Stage 2 cross-validation (25 settings × 4 folds)	37.4
Stage 2 ablation, calibration, refit and 2024 evaluation	24.5
Total	**72.6**

Table 5. Wall-clock time of the notebook's stages on this machine in the submitted run (t40_stage_timings.csv).

4. Data Cleaning

4.1 Data quality audit finding

- **Rainfall values:** No missing timestamps, station IDs or rainfall readings were found. All readings were non-negative and recorded in millimetres (mm). However, one unusually high reading of 37.2 mm in five minutes was identified as a likely instrument error.
- **Timestamp inconsistencies:** Approximately 1.78 million readings from 2017 had timestamps ending in hh:m4:59 instead of the expected five-minute intervals (hh:m5:00). Rounding timestamps to the nearest five-minute interval corrected this issue and revealed 86 duplicate station-time records.
- **Faulty gauges:** Two types of anomalies were identified that standard range checks could not detect: gauges reporting zero rainfall throughout wet months and unusually high daily rainfall compared with nearby stations. Comparison with neighbouring gauges identified 67 potentially faulty station-months and four anomalous spike days, including one reporting 749 mm when nearby stations recorded at most 32 mm.
- **Timestamp interpretation:** The dataset does not specify whether timestamps represent the start or end of a five-minute measurement interval. We assume they indicate the start to avoid using rainfall measured after the prediction time. However, publication delays of 2–10 minutes mean some readings may not be available in real time. Similarly, identifying faulty gauges using daily or monthly patterns requires historical data that would only be available later. These limitations are discussed in Section 6.

4.2 Cleaning steps

Cleaning step	Records remaining	Change
Raw records (bronze)	48,538,713	-
After the range check (one 37.2 mm reading removed)	48,538,712	-1
After de-duplication of station-slots	48,538,626	-86
After removing dead-gauge station-months	48,340,082	-198,544
After removing spike station-days (final, silver)	48,338,938	-1,144

Table 6. Records remaining after each cleaning step. 99.59 % of the raw records are kept.

> The **gauge-fault rules** remove 0.4 % of the readings but affect the conclusions. Without them, S107 would produce an apparent record 147 mm hour in 2022 in Task 1, while the dead months would lower the island-wide means. We flag a station-month as dead when its rain rate is below 10 % of the network median and that median is at least 50 mm. We flag a station-day as a spike when it exceeds 100 mm and 4× the wettest gauge within 5 km.
>
> We leave missing **readings** as unknown rather than filling them **with zeros**. A usable station-day requires 80 % of its slots, and an island-day requires at least 20 usable gauges. We also repeat the analyses on a fixed 21-gauge network to check the effect of changes in coverage.

Duplicate station-time records are removed by retaining the reading with the latest update timestamp. We use Spark's max() aggregation on a structure containing the update time and rainfall value, avoiding the need for a window-based sorting operation across 48 million records. The physical execution plan confirms a single data shuffle, followed by local sorting within partitions using SortAggregate, rather than a global sort.

```
+- SortAggregate(key=[station_id, ts], functions=[max(struct(upd, upd, rain_mm, rain_mm)),
count(1), …
   +- Sort [station_id ASC NULLS FIRST, ts ASC NULLS FIRST], false, 0
      +- Exchange hashpartitioning(station id, ts, 96)
         +- SortAggregate(key=[station id, ts], functions=[partial max(struct(upd, upd, rain mm,
rain_mm)), …
            +- Sort [station_id ASC NULLS FIRST, ts ASC NULLS FIRST], false, 0
```

5. Weather Patterns (Task 1)

We calculate all Task 1 quantities with Spark aggregations of the cleaned readings. Completeness criteria were applied to ensure reliable comparisons across years and minimise the impact of changes in the monitoring network.

5.1 Year to year: dry years have fewer rain events, not lighter rain

Year	Rainfall (mm)	Wet days (≥ 1 mm)	mm per wet day	Events per gauge	Gauges
2017	2,454	239	10.3	403	58
2018	2,656	245	10.9	374	56
2019	1,787	172	10.4	266	54
2020	2,631	230	11.4	365	80
2021	2,900	214	13.6	381	74
2022	2,893	241	12.0	437	72
2023	2,879	228	12.6	412	73
2024	2,708	233	11.6	397	70

Table 7. Annual island-wide rainfall (gap-filled), wet days scaled to a full year, and rain events per gauge.

Annual rainfall ranges from **1,787 mm** in **2019 to 2,900 mm** in **2021**, with a mean of 2,613 mm. The dry year of 2019 has 172 wet days, compared with 233 in the other years, while its 10.4 mm per wet day is close to the usual 11.8 mm. This points to less frequent rain as the main difference. The dry condition were also prolonged, 2019 had 12 consecutive month of below average rainfall.

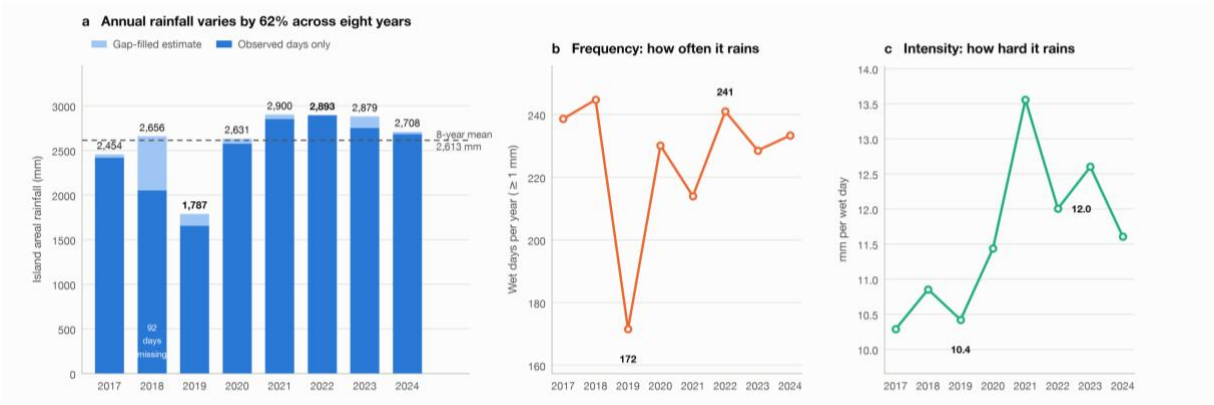


Fig 1. Annual rainfall (a) split into frequency (b, wet days) and intensity (c, mm per wet day).

5.2 Within a year: two wet seasons and two daily rhythms

Monthly rainfall shows two wetter periods, with higher rainfall in Nov (315 mm), Dec (261 mm), and Apr (248 mm), while Feb (145 mm) and Jul (147 mm) are the driest months. Daily rainfall patterns also vary by season. From June to September, rainfall peaks around 10:00 AM, consistent with morning squalls, while other months show afternoon peaks around 3:00–4:00 PM, likely due to convective rainfall. Rainfall is highly concentrated: the wettest 10% of rainy five-minute intervals account for 49% of total rainfall, while the wettest 1% contribute 10%. Most rainfall events are short and light, with a median duration of 20 minutes and an accumulation of 1.0 mm.

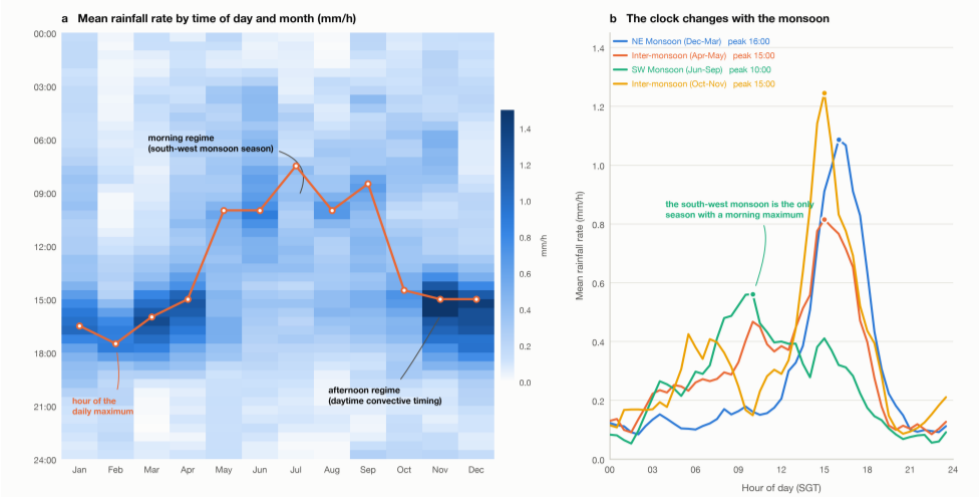


Fig 2. Mean rainfall rate by half-hour and month (a) and by season (b): a 10:00 peak in the south-west monsoon, a 15:00–16:00 peak otherwise.

5.3 Across the island: The West is Twice as Wet as the East

Rainfall varies considerably across Singapore. Between 2021 and 2024, average annual rainfall across 53 stations ranged from 3,548 mm (S228) to 1,700 mm (S208), decreasing by approximately 266 mm per 10 km eastward ($r = -0.68$). Rainfall is also highly localised: the correlation between 30-minute rainfall totals falls from 0.84 for stations within 2 km to 0.15 beyond 20 km, with a characteristic correlation distance of approximately 8.5 km. These findings support using station-specific predictions and neighbouring rainfall observations within 10 km in Task 2.

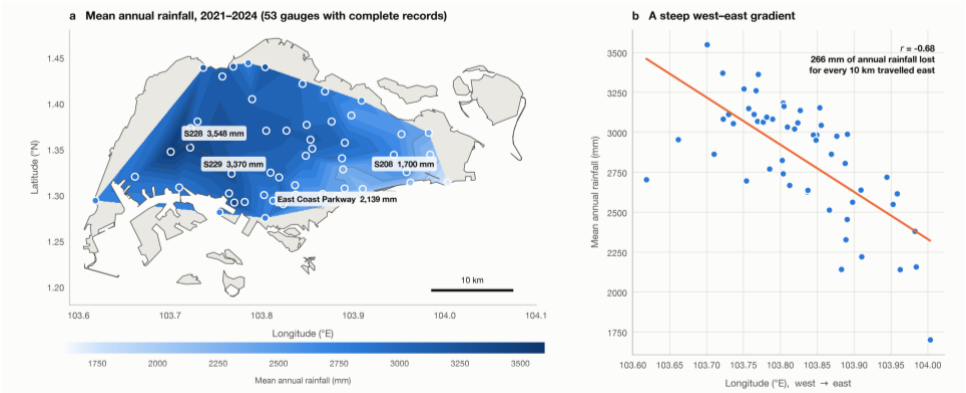


Fig 3. Mean annual rainfall 2021–2024 interpolated between gauges on Singapore's land area (a), and its west–east gradient (b).

5.4 Heavy rainfall: increased frequency but no statistically significant trend

Heavy rainfall events (≥ 5 mm in 30 minutes) are **about 23 % more frequent in 2020–2024 than in 2017–2019**. The increase is 20 % on the fixed 21-gauge network, so the 2020 network expansion explains little of it. However, exact Mann-Kendall tests on the eight annual values find **no statistically significant trend in any series** (Table 8). This short record cannot clearly separate a trend from year-to-year variability; the tests do not establish that rainfall is unchanged.

Annual series	Kendall τ	p-value	Sen slope (% of mean per year)
annual rainfall, gap-filled (mm)	+0.36	0.28	+2.3 %
wet days per year (≥ 1 mm, scaled)	-0.07	0.90	-0.3 %
intensity (mm per wet day)	+0.57	0.06	+2.7 %
rain events per gauge	+0.29	0.40	+1.6 %
30-min totals ≥ 5 mm, rate (all gauges)	+0.43	0.18	+2.4 %
30-min totals ≥ 5 mm, rate (fixed network)	+0.36	0.28	+1.4 %
annual max 30-min depth (mm, any gauge)	-0.36	0.28	-1.2 %
annual max 60-min depth (mm, any gauge)	+0.29	0.40	+1.5 %

Table 8. Mann-Kendall trend tests on the eight annual values (exact two-sided p-value) with Sen's slope.

Robustness check: Annual and monthly results from 21 stations reporting throughout 2017–2024 closely match the full network: annual totals correlate at r = 0.986, the driest year and wettest month are unchanged, and monthly anomalies agree in sign 97.9% of the time.

Implications for Task 2: Recent rain, calendar features, station location and gauges within 10 km capture the temporal and spatial patterns. Uneven rainfall and the 0.83% event rate at 5 mm motivate exceedance probabilities and metrics suited to imbalanced data.

6. Feature Engineering

Features were generated using Spark window functions on each station's time series. Time-based windows (rangeBetween) calculate rainfall features over fixed intervals, ensuring that missing readings are not replaced by older observations. We also count available readings to identify incomplete windows. Neighbouring station features were calculated using a small broadcast table of station pairs based on their coordinates, joined with rainfall readings and grouped by target station and time slot. This avoids expensive station-by-station joins across the 48.5 million-record dataset.

Key design principle - Preventing Data Leakage: The model uses station identifiers and coordinates, time-related features, and rainfall readings timestamped at or before t − 5 minutes, ensuring that no future rainfall measurements are used as features. To validate this, we independently recalculated the features and targets using pandas. We checked 300 randomly selected observations for Stage 1 and, for Stage 2, 300 observations for station-level features and targets and 30 for neighbouring and island-wide features. All results matched the Spark calculations after correcting an issue in Stage 2 where neighbouring station counts included gauges that began reporting later in the day.
Real-Time Limitations: These checks confirm that the feature calculations do not use future rainfall measurements, but they do not guarantee real-time availability. Rainfall readings may be published after their timestamps, and some gauge-quality checks rely on complete daily or monthly observations. These limitations are discussed further in the Limitations section.

Feature Selection - Task 1 guides a compact baseline feature set in Stage 1 and additional rainfall and spatial features in Stage 2. Cross-validation compares the top 8, 16 and all 33 candidates, selecting all 33.

Group	Features	Rationale (Task 1 findings)
Model Inputs	Rainfall threshold (h): log(1 + h) and its interactions with recent rainfall features in Stage 1; both h and log(h) in Stage 2. Station identifier.	Required inputs for station-specific rainfall prediction.
Time	Sine and cosine transformations of time of day and day of year; season.	Rainfall patterns vary by time of day and monsoon season.
Location	Station latitude and longitude.	Rainfall generally decreases from west to east, with the wettest gauge recording approximately twice as much rainfall as the driest.
Recent Rainfall	Rainfall in the last 5, 15, 30, 60 and 180 minutes; number of wet readings and peak rainfall in the last hour. Stage 2 only: Recent rainfall trend, time since last rainfall, number of available readings and rainfall accumulated today.	Rainfall events typically last around 20 minutes, and wet or dry conditions often persist.
Neighbouring gauges	Stage 1: Rainfall statistics from up to four stations within 10 km. Stage 2: Average and maximum rainfall, proportion of wet stations and data availability within 5 km, within 10 km, and 5–20 km to the west.	Nearby stations show strong rainfall correlation over distances of approximately 8.5 km, and squall lines often approach from the west.
Island-wide Rainfall (Stage 2 only)	Proportion of reporting stations experiencing rainfall and their average rainfall over the previous 30 minutes.	Captures broader wet and dry conditions across Singapore.

Table 9. Feature groups. Stage 1 uses the compact set; Stage 2 adds the richer items and lets cross-validation choose.

7. Modelling Approach

7.1 Chronological split and year-blocked cross-validation

No random splits. We use chronological data splits rather than random splits to reduce data leakage, as consecutive five-minute observations share five of the six readings used to calculate rainfall over the next 30 minutes. Stage 1 trains on 2017-2022, uses 2023 for model selection, and evaluates performance on 2024. Stage 2 uses Spark's CrossValidator with four year-blocked folds (2017-2018, 2019-2020, 2021-2022, and 2023), followed by retraining on 2017-2023 and final evaluation on 2024. The 2024 test year is excluded from model development, with the boundary purged to prevent overlapping target windows (Section 8.1).
However, the boundaries between cross-validation folds are not purged. Approximately 369 of the 13.7 million prediction rows have target windows extending up to 20 minutes into the next fold. While this introduces a small degree of overlap between folds, it affects only a negligible proportion of observations.

7.2 Training sets

The two stages use different data selection and sampling strategies to improve computational efficiency while maintaining reliable probability estimates.
Stage 1: Observations must have a complete three-hour rainfall history and all six future five-minute readings, resulting in 32,962,257 eligible prediction times from 2017–2022. For each of 11 rainfall thresholds (0.2–30 mm), observations are sampled using a hash of station, time and threshold. Sampling aims for approximately 8,000 positive examples per threshold, with a maximum sampling rate of 6%. Since sampling is independent of the rainfall outcome, the original event rates are preserved in expectation, and no weighting is required.
Stage 2: Observations with incomplete rainfall histories are also retained, using the number of available readings as a feature. However, all six future readings are still required to calculate the target. To reduce redundancy, prediction times are sampled every 15 minutes during 2017–2023, while the 2024 test data retains five-minute intervals. Stage 2 retains all observations with rainfall in the previous three hours or the next 30 minutes, but only 3% of the remaining quiet observations. As this over-represents rainy periods, inverse inclusion weights are applied: 1 for always-retained observations and approximately 33.3 for sampled quiet observations. These weights are used in Spark MLlib training and cross-validation through weightCol. Each selected observation is then paired with three thresholds sampled independently of the outcome from a log-uniform distribution between 0.2 and 60 mm. The Stage 2 2024 test set is neither sampled nor weighted, ensuring that final evaluation reflects the original distribution of eligible observations.

7.3 Models evaluated

Candidate	PR-AUC	ROC-AUC	Log-loss skill	Threshold violations
Climatology (season × hour × threshold)	0.043	0.720	0.000	-
Base rate per threshold	0.018	0.500	-0.059	-
Logistic: own gauge only	0.403	0.899	0.325	0.00 %
Logistic: + 4 neighbouring gauges	0.445	0.929	0.373	0.00 %
Logistic: + threshold interactions	0.458	0.929	0.393	0.00 %
Logistic: + interactions, stronger L2	0.463	0.930	0.383	0.00 %
GBT, depth 5 (80 trees)	0.452	0.925	0.029	0.03 %
GBT, depth 7 (80 trees)	0.449	0.933	0.031	0.22 %

Table 10. Stage 1 model performance on 2023 validation data, averaged equally across six rainfall thresholds. Models were trained on 2017–2022 using Spark ML pipelines. The selection criterion was the highest log-loss skill.

Adding neighbouring station features improved logistic regression PR-AUC from 0.403 to 0.445, while threshold interactions further improved log-loss skill to 0.393, the highest among Stage 1 candidates. Although gradient-boosted trees (GBT) achieved comparable ranking performance, their low log-loss skill (0.029–0.031) indicated poor probability estimates. Logistic regression with threshold interactions was therefore selected as the Stage 1 baseline. Stage 2 further investigates tree-based models with probability calibration.

Model family	Settings tried	Best setting	CV log loss (mean ± sd over folds)
Logistic regression	6	regParam=0.001, elasticNetParam=0	0.0391 ± 0.0022
Random forest	4	numTrees=100, maxDepth=12	0.0319 ± 0.0018
Gradient-boosted trees	12	maxDepth=4, maxIter=100, stepSize=0.1	0.0403 ± 0.0018

Table 11. Stage 2: best cross-validated setting per model family (4 year-blocked folds, weighted log loss, 22 settings in total). The selector size k was tuned first with logistic regression (k=33: 0.0425, k=16: 0.0454, k=8: 0.0585); all 33 candidates were kept.

Cross-validation selects all 33 features and the random forest with 100 trees and maximum depth 12, which has the lowest log loss (0.0319). Section 7.4 covers calibration and consistency.

7.4 Calibration and consistency

Tree-based models do not automatically produce well-calibrated probabilities. To improve reliability, we applied Platt-style calibration using logistic regression on the logit of the predicted probability, the log of the rainfall threshold, and their interaction. The calibrator was fitted using 2023 predictions from models trained on 2017-2022. It was trained on the first half of 2023 and validated on the second half, retaining it only if it reduced log loss. The selected calibrator was then refitted on the full 2023 dataset. Calibration reduced held-out log loss from 0.0294 to 0.0288 for random forest and from 0.0381 to 0.0311 for GBT.

Probability Consistency: Tree-based models do not guarantee that predicted probabilities decrease as rainfall thresholds increase. We therefore applied a running-minimum correction across the six evaluated thresholds for each station and prediction time. For the final random forest, this affected at most 0.055% of 2024 predictions.

The final predict() function applies the same correction to a fixed grid that includes the six evaluated thresholds, then interpolates on the log-threshold scale to estimate probabilities for other thresholds within the supported range (0.2–60 mm). This ensures that predictions remain consistent with the evaluation procedure and do not increase as rainfall thresholds rise.

The Stage 1 logistic regression required no correction, with zero monotonicity violations across approximately 25.8 million adjacent threshold pairs evaluated in 2023 and 2024.

8. Results

8.1 Final random forest on 2024 test data

The final random forest was retrained on 2017–2023 data, excluding observations whose target windows extended into 2024 to prevent data leakage. It was evaluated on all 6,371,188 eligible prediction times in 2024 across six rainfall thresholds, without sampling or weighting. Precision, recall and F1 were evaluated using cutoffs selected on 2023 validation data.

h (mm)	Event rate	PR-AUC	ROC-AUC	Log loss	Brier skill	Cut-off	Precision	Recall	F1
0.5	3.63 %	0.721	0.965	0.0645	0.492	0.40	0.71	0.59	0.65
1	2.57 %	0.667	0.966	0.0516	0.448	0.35	0.65	0.58	0.61
2	1.69 %	0.587	0.966	0.0395	0.386	0.25	0.57	0.54	0.55
5	0.84 %	0.465	0.969	0.0240	0.293	0.20	0.48	0.46	0.47
10	0.38 %	0.351	0.973	0.0130	0.211	0.15	0.42	0.38	0.40
20	0.10 %	0.185	0.982	0.0045	0.101	0.10	0.26	0.28	0.27

Table 12. Final model on the 2024 test year by threshold. Brier skill = share of the climatology error removed.

At the 5 mm threshold, where rainfall events occur in only 0.84% of observations, the model achieves a PR-AUC of 0.465 (approximately 55 times the event rate) and ROC-AUC of 0.969. It also reduces the climatology baseline's squared probability error by 29.3%. Using the cutoff selected on 2023 data, the model identifies 46% of actual events with 48% precision.

Predictive performance declines for rarer events, with PR-AUC decreasing from 0.721 at 0.5 mm to 0.185 at 20 mm, although ROC-AUC remains above 0.96 across all thresholds. At 5 mm, probability calibration is generally good, with a low expected calibration error (0.0016), although some probabilities are under- or overestimated.

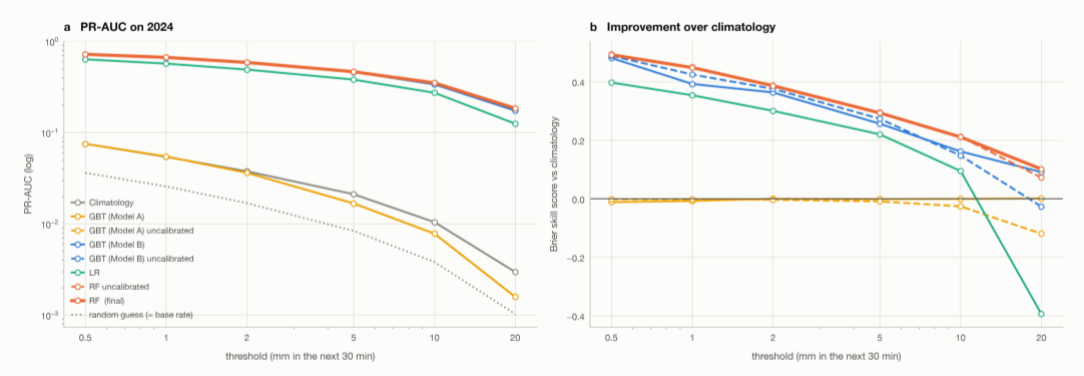


Fig 4. PR-AUC (a) and Brier skill against climatology (b) on all of 2024 for every Stage 2 model. Model A, which sees no rain observation, is no better than climatology.

8.2 Model families at 5 mm

Model	PR-AUC	ROC-AUC	Log loss	Brier skill
Climatology (station × month × hour)	0.021	0.679	0.0483	0.000
GBT, Model A: threshold + time + station only	0.017	0.684	0.0473	-0.002
Logistic regression (tuned)	0.381	0.951	0.0289	0.220
GBT, Model B (tuned, calibrated)	0.460	0.966	0.0263	0.256
Random forest, uncalibrated	0.465	0.969	0.0238	0.295
Random forest, calibrated	0.465	0.969	0.0240	0.293

Table 13. Stage 2 models on all 2024 rows at the 5 mm threshold. Model A has only the threshold, time and station.

8.3 Head-to-head on identical rows

We compare both stages on the same 1,255,003 observations per threshold from the 2024 test year, matched by station, prediction time and rainfall threshold. The final random forest improves PR-AUC, ROC-AUC and log loss at all five thresholds. Its richer inputs include recent rainfall and neighbouring-station observations.

h (mm)	Events	PR-AUC clim.	PR-AUC Stage 1	PR-AUC final	ROC-AUC Stage 1	ROC-AUC final	Log loss Stage 1	Log loss final
1	32,509	0.049	0.585	0.663	0.918	0.967	0.0661	0.0521
2	21,290	0.033	0.524	0.587	0.915	0.968	0.0489	0.0395
5	10,445	0.018	0.418	0.459	0.912	0.970	0.0287	0.0238
10	4,790	0.010	0.327	0.352	0.917	0.975	0.0150	0.0129
20	1,323	0.004	0.186	0.191	0.941	0.982	0.0050	0.0046

Table 14. Stage 1 logistic model vs the final model on identical 2024 rows.

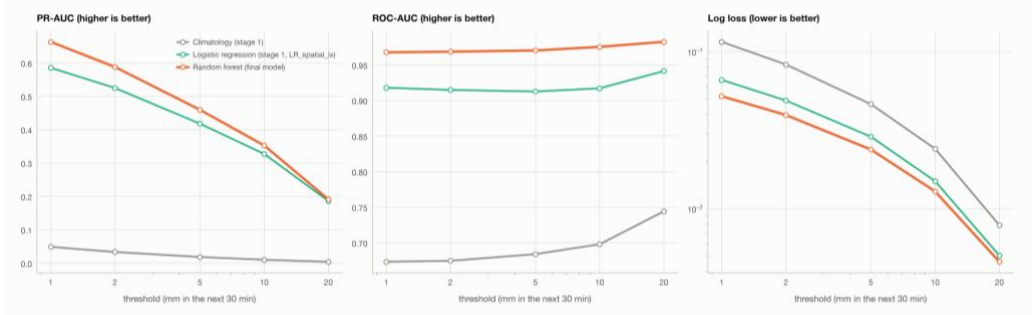


Fig 5. PR-AUC, ROC-AUC and log loss by threshold for the climatology, the Stage 1 logistic model and the final model on identical 2024 rows.

8.4 Strong and weak scaling on the laptop

We evaluated Spark's scalability by benchmarking the main cleaning job: reading Parquet data, correcting timestamps, removing duplicates and aggregating rainfall into station-day records. Each configuration ran in a fresh Spark session, with execution time measured as the median of three runs after a warm-up. **Strong Scaling:** Using all eight years of data, increasing from 2 to 16 cores reduced execution time from 27.8 to 5.7 seconds, achieving a

4.88× speed-up and 61% parallel efficiency. However, gains diminished at higher core counts. An Amdahl's law fit estimated a serial component of 3.4%, suggesting a theoretical maximum speed-up of approximately 30× relative to one core. **Weak Scaling:** As the dataset increased from 5.3 million to 48.5 million records and cores increased from 2 to 16, execution time rose from 3.3 to 5.8 seconds. This achieved a 5.25× scaled speed-up and 66% efficiency, demonstrating Spark's ability to handle larger datasets with relatively small increases in execution time.

Cores	Wall time (s)	Speed-up vs 2 cores	Amdahl fit	Ideal	Parallel efficiency
2	27.8	1.00×	1.00×	1×	100 %
4	13.6	2.04×	1.88×	2×	102 %
8	8.0	3.48×	3.34×	4×	87 %
16	5.7	4.88×	5.49×	8×	61 %

Table 15. Strong scaling: all 8 years on 2–16 cores (median of three runs; t42_scaling.csv, t47_amdahl_fit.csv).

Cores	Years	Rows (M)	Wall time (s)	Scaled speed-up	Ideal	Efficiency
2	2017	5.3	3.3	1.00×	1×	100 %
4	2017-2018	9.9	3.5	1.77×	2×	88 %
8	2017-2020	21.2	3.9	3.41×	4×	85 %
16	2017-2024	48.5	5.8	5.25×	8×	66 %

Table 16. Weak scaling: the data grows with the cores (median of three runs; t45_weak_scaling.csv). Efficiency = throughput per core relative to 2 cores.

Both experiments show declining efficiency at higher core counts, partly due to shared memory and storage resources and the laptop's mix of performance and efficiency cores. Distributed clusters can reduce these hardware constraints by providing dedicated resources on each worker node (Section 8.5).

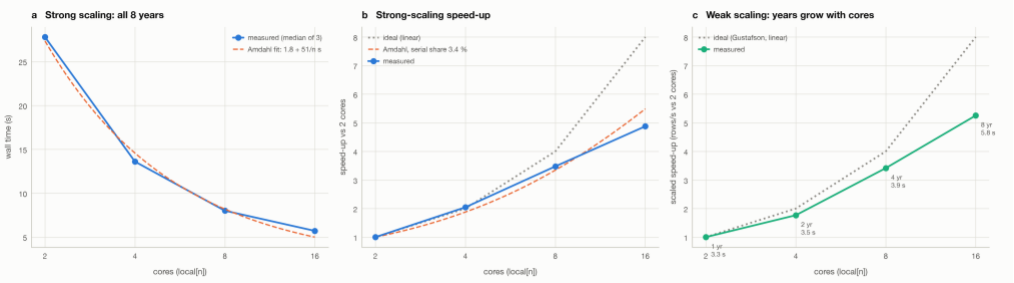


Fig 6. Strong scaling: wall time with the Amdahl fit (a) and speed-up (b); weak scaling: scaled speed-up when the years grow with the cores (c).

8.5 The same job on a Google Cloud Dataproc cluster

The same job ran on a Dataproc cluster with 2 and then 4 n2-standard-2 workers, reading the bronze layer from Cloud Storage. Both runs returned the same answer as the laptop. Ingesting the 7.5 GB of CSV from the bucket into bronze Parquet took 328 s on 2 workers. Doubling **the workers** reduced cleaning **time from 74.8 s to 37.3 s, a 2.0× speed-up**. The laptop gained 1.7× when moving from 4 to 8 cores (13.6 s → 8.0 s).

Platform	Configuration	Cores	Wall time (s)	Speed-up	Efficiency
Laptop	local[2]	2	27.8	1.00×	100 %
Laptop	local[4]	4	13.6	2.04×	102 %
Laptop	local[8]	8	8.0	3.48×	87 %
Laptop	local[16]	16	5.7	4.88×	61 %
Dataproc cluster	2 workers x n2-standard-2	4	74.8 (73.4-76.1)	1.00×	100 %
Dataproc cluster	4 workers x n2-standard-2	8	37.3 (33.9-43.7)	2.01×	100 %

Table 17. The cleaning job on the laptop and on the Dataproc cluster (cluster: mean of three timed runs, range in brackets; t44_local_vs_cloud.csv).

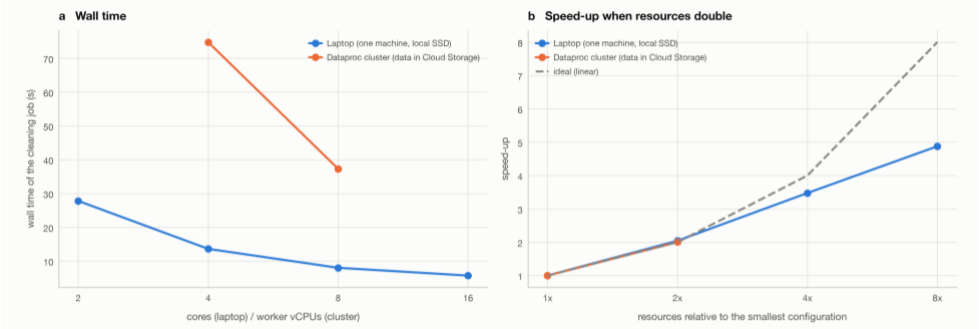


Fig 7. Wall time (a) and speed-up when resources double (b): laptop cores against Dataproc workers for the same job.

At the same core count, the laptop runs this job about 5× faster than the cluster. It has fast cores, 128 GB of memory and a local SSD. Each n2-standard-2 worker has 2 vCPUs, reads the input over the network from Cloud Storage (requests took 0.3-0.5 s in the job logs) and spills shuffles to standard disks. Adding laptop cores shares the same memory bus and disk, and efficiency falls to 87 % at 8 cores and 61 % at 16. Adding workers also adds CPU, memory and disk, consistent with the almost proportional speed-up from 2 to 4 workers.

For this 7.5 GB workload, a strong single machine is the fastest and cheapest option. A cluster becomes useful when the data or model outgrows one machine; the near-linear scaling observed here supports adding workers as the workload grows. The education project's 12-vCPU quota limited this experiment to 4 workers.

8.6 The complete pipeline on the Dataproc cluster

We also ran the **whole notebook** on a Dataproc cluster, including cleaning, Task 1, both stages of Task 2 with full cross-validation, the 2024 evaluation and predict(). The cluster had a master and 2 n2-highmem-4 workers (4 vCPU, 32 GB; the 12-vCPU quota), with Spark 3.5.3 on YARN and the data layers on HDFS. This was the notebook **as it stood on 14 October 2026**, converted by cloud/make_dataproc_notebook.py, which changes only the Spark master, storage and laptop core benchmark. It predates four later revisions: the Spark-plan cell, the strong/weak-scaling rewrite, the Stage 2 train/test boundary purge and the fixed-grid predict(). The executed copy and a note on these differences are in cloud/full_run/. The run completed without errors in 290 minutes of timed work, compared with 73 on the laptop, or 4.0× slower. As in Section 8.5, the cluster used 8 executor cores on 2.8 GHz Xeons with shuffles crossing the network, compared with 14 fast laptop cores sharing one memory.

Stage	Laptop (min)	Cluster (min)	Cluster / laptop
Ingestion and data-quality audit	0.4	2.6	6.7×
Cleaning and pre-processing	0.6	2.5	4.0×
Task 1 analyses	0.5	2.5	4.6×
Stage 1: features, sets, baselines and models	7.9	30.2	3.8×
Stage 2: features and screening	1.2	5.1	4.1×
Stage 2: cross-validation (100 fits)	37.4	161.9	4.3×
Stage 2: ablation, calibration, refit, 2024 test, head-to-head	24.5	84.8	3.5×
Total	**73**	**290**	**4.0×**

Table 18. The same notebook on the laptop and on the Dataproc cluster: minutes per stage (t40_stage_timings.csv of each run).

Result	Laptop	Cluster
Cleaning, Task 1 tables and Stage 1 data sets (9 tables)	identical	identical (largest difference 1e-12)
Stage 1 logistic model: 2024 macro PR-AUC	0.4462	0.4462
Cross-validation: best setting of all four grids	-	same settings
Cross-validation: random-forest log loss	0.0319	0.0319
Final random forest on 2024: PR-AUC at 1 / 5 / 20 mm	0.667 / 0.465 / 0.185	0.666 / 0.465 / 0.189
Head-to-head at 5 mm: final vs Stage 1 PR-AUC	0.459 vs 0.418	0.460 vs 0.418

Table 19. Results of the two runs. Everything deterministic is identical; the tree models differ only in the third decimal because their row subsampling depends on partitioning.

The only Task 1 difference is the 99th-percentile event depth, which changes by 0.26 mm. It uses Spark's approximate percentile, whose result depends on data partitioning. The archived cluster run reproduced the pipeline's conclusions on a YARN cluster, providing evidence that the code is portable across the two environments.

8.7 Verification

- Metric validation: Independent PR-AUC (0.4448) and ROC-AUC (0.9268) calculations were consistent with Spark MLlib results (0.4417 and 0.9269).
- Data leakage: Independent pandas calculations matched Spark's features and targets. Both stages excluded incomplete future targets and prevented overlap with the 2024 test year.
- Reproducibility: Task 1 and logistic regression results were consistent across runs, while tree-based models showed minor variations due to partition-dependent sampling.

9. Interpretation & Insights

9.1 What drives the forecast

Feature	Group	Importance	Explanation
thr	mandatory input	0.193	The threshold itself: the probability falls steeply as the asked-for amount rises.
log_thr	mandatory input	0.167	Log of the threshold, the same input on a scale the trees split on more easily.
r5	recent rain	0.140	Rain at the station in the last 5 minutes. Rain now is the best predictor of rain in the next half-hour.
min_since_rain	recent rain	0.074	Minutes since the gauge last saw rain (capped at 24 h). A long dry spell makes rain in the next 30 minutes unlikely.
r15	recent rain	0.068	Rain in the last 15 minutes: is it a passing shower or a storm?
outer_mean	neighbours	0.065	Mean 30-minute rain at gauges within 10 km: is a storm nearby?
r30	recent rain	0.040	Rain in the last 30 minutes.
inner_mean	neighbours	0.037	Mean 30-minute rain at gauges within 5 km.
outer_max	neighbours	0.034	The wettest gauge within 10 km.
inner_max	neighbours	0.032	The wettest gauge within 5 km.

Table 20. Most important features of the final model (mean decrease in impurity) with a plain-English reading.

The feature-group ablation supports this interpretation. We retrained the GBT without each group and validated it on 2023. Removing **recent rain** at the station raises log loss by +7.8 %, removing **neighbours** by +2.9 % and removing the **island-wide** signal by +0.7 %. Removing time costs +0.2 %, while removing latitude/longitude has no effect because the station index already carries location. Model A, which uses no rain observations, is +87.3 % worse.

9.2 Where the model works and where it struggles

- At 5 mm, Brier skill is highest **in the Inter-monsoon (Oct-Nov)**, at 0.33, and lowest in the NE Monsoon (Dec-Mar), at 0.27. Differences in storm type may contribute to this seasonal variation.
- Brier skill is lowest **around 12:00** (0.11) and highest at 23:00 (0.48). One possible explanation is that afternoon convective storms often begin with little warning in the gauge record, whereas night-time rain may already be visible at other gauges.
- **Heavy thresholds are** harder to predict. At 20 mm, exceeded in 0.1 % of half-hours, PR-AUC is 0.185. ROC-AUC remains high, so the model still ranks risky half-hours well, although it cannot determine how heavy an individual storm will be.
- A storm can stop or start without warning, so an individual forecast can miss. The reliability diagrams assess the probabilities across many forecasts.

9.3 Spark concepts from the course in this pipeline

Narrow transformations work within separate partitions. **Wide transformations** bring rows with the same key together by shuffling data between executors. These wide steps account for much of the pipeline's cost, so we designed them to limit the data shuffled:

Step	Spark operation	Type	Data movement
Typing, timestamp snap, range check, reading type	select, withColumn, filter	narrow	none
De-duplicate station-slots	groupBy(station, slot).agg(max(struct(update, value)))	wide	one shuffle; sort-based aggregate within partitions (no global sort)
Station-days, island-days, monthly and annual totals	groupBy aggregations	wide	partial aggregation, then a shuffle of small partial results
Rolling rain windows, time since rain, storm trend	Window.partitionBy(station).orderBy(time).rangeBetween	wide	shuffle by station, sort within partitions
Neighbour rings and gauge-fault checks	join with a broadcast station-pair table	narrow on the large side	the small table is copied; the 48 M rows stay put
Station features × neighbour features	join on (station, slot)	wide	sort-merge join: both sides shuffled
Hash sampling, threshold expansion	filter, explode	narrow	none

Table 21. Narrow and wide transformations in the pipeline.

For the joins in this pipeline, Spark broadcasts the smaller side when it is below the 10 MB broadcast threshold and otherwise uses a sort-merge join. The physical plans in Section 12.4 of the notebook show a BroadcastHashJoin for the station-pair table, both with and without our hint. Joining the two large feature tables uses a **SortMergeJoin** and shuffles both sides. **Adaptive query execution** revises the plan at runtime. After the grouped aggregation runs, its final plan shows AQEShuffleRead coalesced merging the 64 post-shuffle partitions because fewer partitions are needed for the actual data.

```
-- small x large, F.broadcast hint (neighbour rings):  +- BroadcastHashJoin [nb_id], [nb_id], Inner, BuildRight, false
-- small x large, no hint (Spark decides):  +- BroadcastHashJoin [nb_id], [nb_id], Inner, BuildRight, false
-- large x large (station features x neighbour features):  +- SortMergeJoin [station_id, slot_epoch], [station_id, slot_epoch], LeftOuter
-- grouped aggregation after it ran (AQE final plan):  *(2) HashAggregate(keys=[year], functions=[sum(rain mm)], output=[year, mm])
```

MLlib fits the logistic regression with L-BFGS. At each iteration, every partition computes the loss and gradient for its own rows, and treeAggregate combines the partial results before the driver takes a step. Random forest and GBT use feature binning (maxBins=128) and distributed aggregation of split statistics to construct decision trees. Spark's CrossValidator allows up to four model fits to run concurrently (parallelism=4) across year-blocked folds, while sampling weights are incorporated through weightCol.
For data processing, we use columnar Parquet storage partitioned by year, enabling Spark to skip irrelevant partitions during queries and scalability experiments. Although Spot workers, local SSDs and autoscaling could improve cluster efficiency, resource quotas limited their use. Autoscaling was also less suitable for our HDFS-based setup, where removing workers could affect data availability.

10. Limitations, Risks & Future Work

Limitation	Detail
One test year	2024 is the only untouched year; eight years are too short to separate climate trends from variability.
Rare heavy rain	Only a few hundred training positives above 30 mm, so probabilities for very heavy thresholds are the least certain.
Gauge data only	The model sees only rain already falling on a gauge. Radar, satellite or wind fields would show storms before they reach any gauge.
Grid edges	Some best settings (RF trees and depth, GBT depth and iterations) lie on the edge of their search grid; a wider grid could improve them further.
Cluster size	The 12-vCPU quota limited the cluster to 2–4 small workers, so the cluster ran slower than the laptop; the submitted model is the laptop-trained one (the complete cluster run reproduced its results).
Timestamp meaning	The data documentation does not say whether a stamp marks the start or end of its interval; we chose the reading under which no future information can leak, at worst labelling the prediction time five minutes late.
Real-time deployment	Rainfall readings may be published 2–10 minutes after their timestamps, and some gauge-quality checks require complete daily or monthly data. A real-time system would need a longer observation lag and quality checks using only information available at prediction time.

Table 22. Data and model limitations.

Further **work** could add radar or satellite inputs and extend tuning beyond the grid edges. Model training could run on a larger Dataproc cluster with local SSDs. The RAPIDS accelerator for Apache Spark is another option: on a GPU cluster, it runs filters, projections, sorts and joins in the GPU's columnar memory. Seasonal recalibration could also help account for changes in the gauge network.

11. Conclusion

This project used Apache Spark to analyse 7.5 GB of historical rainfall data, applying data cleaning and quality checks to ensure reliable results. Task 1 identified two wet seasons, two distinct daily rainfall patterns, a west-east rainfall gradient and fewer rain events in dry years. These findings guided feature selection for Task 2, where year-blocked cross-validation selected a random forest model to estimate rainfall exceedance probabilities. On the unseen 2024 test data, the model achieved a PR-AUC of 0.465 at the 5 mm threshold, outperforming the logistic regression baseline across all evaluated thresholds and producing generally well-calibrated probabilities.

The key takeaway is that data quality and feature engineering are critical to reliable predictions. Detecting faulty gauges prevented misleading results, while validation checks identified and corrected data leakage. This highlights the importance of careful data preparation and feature selection alongside model choice.

Declaration of AI usage - Generative AI tools (Claude by Anthropic, and ChatGPT by OpenAI) were used to help write and debug the PySpark code, discuss design choices, and draft and edit parts of this report. All numbers come from the submitted notebook or, for Sections 8.5–8.6, from the archived Dataproc runs. The authors reviewed all code, results and text and take full responsibility for the submission.